

Paper VII: The Information-Acoustic Eversion

Sphere Eversion as Hecke Propagation through the $\text{Cl}(3,1)$ Substrate

Kristin Tynski

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Abstract

We show that sphere eversion—the orientation reversal of an immersed S^2 in $\mathbb{R}^3$ —admits a natural arithmetic realization as Hecke propagation on the 4-dimensional Hilbert modular homology $H_1(X_0((11)), \text{cusps}; \mathbb{Z})$ over $K = \mathbb{Q}(\sqrt{5})$. A coherence state $\psi \in H_1 \cong \mathbb{R}^4$ evolves under $\psi_k = T_{\mathfrak{p}_k} \psi_{k-1}$, where $T_{\mathfrak{p}} = a_{\mathfrak{p}} \Pi_c + (N(\mathfrak{p})+1) \Pi_e$ is the Hecke operator from Paper VI. The eversion completes when $\prod a_{\mathfrak{p}_k} < 0$ —exactly when the cuspidal component reverses sign. The Thurston corrugation frequencies $k_{\mathfrak{p}} = N(\mathfrak{p})$ and amplitudes $A_{\mathfrak{p}} = |a_{\mathfrak{p}}|/N(\mathfrak{p})$ arise naturally from the arithmetic. A 3-prime sequence ($p = 5, p = 3, p = 31$) achieves a complete eversion with multi-frequency corrugation, verified by 29 tests covering path dynamics, sign flip, rho bridge integrity, acoustic propagation, marching cubes isosurface extraction, and Smale invariant.

What this paper does.

- Realises sphere eversion – the orientation reversal of an immersed S^2 in $\mathbb{R}^3$ – as Hecke propagation on the 4-dim Hilbert modular homology H_1 of Paper VI: $\psi_k = T_{\mathfrak{p}_k} \psi_{k-1}$ on a $\text{Cl}(3,1)$ -coherence state.
- Shows the eversion *completes* exactly when the running cuspidal product $\prod a_{\mathfrak{p}_k}$ becomes negative – the sign flip of the cuspidal component. A parsimonious three-prime sequence ($p = 5, p = 3, p = 31$) achieves a full eversion with three distinct Thurston-corrugation frequencies $k_{\mathfrak{p}} \in \{5, 9, 31\}$.
- Identifies the Thurston corrugation parameters $k_{\mathfrak{p}} = N(\mathfrak{p})$, $A_{\mathfrak{p}} = |a_{\mathfrak{p}}|/N(\mathfrak{p})$ as arithmetic invariants of the prime, not external knobs; verified by 29 tests on path dynamics, sign flip, rho-bridge integrity, and acoustic propagation.

Place in the series. This is *Paper VII* of the Golden Substrate series. Prerequisites: *Paper I (Substrate)* and especially *Paper VI (Operator bridge)*. Independent of Papers III, IV, V, and VIII.

The arc this paper sits in. Sphere eversion is one of the deepest visual results of 20th-century topology – the Smale–Thurston construction that turns the sphere inside out without tearing, without singularities, and without ever fully creasing. Thurston’s explicit construction proceeds by *corrugating* the sphere at ever higher spatial frequencies; in the limit, the corrugation density becomes C^1 -dense, the immersion becomes regularly homotopic to its mirror, and the topological obstruction is defeated. The construction has always felt geometric – the corrugation frequency k and amplitude A are continuous knobs that the topologist adjusts by hand to achieve smoothness.

This paper makes a precise claim: those continuous knobs are not free. They are arithmetic. Under the $\rho : \text{Cl}(3,1) \rightarrow \text{End}(H_1)$ representation of Paper VI, a Hecke operator $T_{\mathfrak{p}}$ at a prime

ideal $\mathfrak{P}$ of norm $N(\mathfrak{P}) = p^r$ acts on a coherence state as a multivector multiplication that, at the level of the volumetric scalar field, produces exactly a Thurston corrugation of frequency $k_{\mathfrak{P}} = N(\mathfrak{P})$ and amplitude $A_{\mathfrak{P}} = |a_{\mathfrak{P}}|/N(\mathfrak{P})$ – with $a_{\mathfrak{P}}$ the cuspidal Hecke eigenvalue, hence determined by the splitting type of p in $\mathbb{Z}[\varphi]$. The continuous knobs are forced to take discrete, arithmetically determined values.

The consequence is parsimony. A single Hecke step achieves one corrugation frequency. A three-step sequence with one prime of each splitting type (ramified, inert, split) achieves three distinct frequencies, and – by the arithmetic of the Hecke eigenvalues – also achieves the sign flip $\prod a_{\mathfrak{P}_k} < 0$ that is the arithmetic signature of a completed eversion. The minimal such sequence is $(\mathfrak{P}_5, \mathfrak{P}_3, \mathfrak{P}_{31})$; it is the eversion in three primes.

We treat the Thurston-eversion / Hecke-on- H_1 identification as a structural claim about the substrate, not a metaphor. The identification is supported by a 29-test verification suite over path dynamics, sign flip, rho-bridge integrity, acoustic propagation, marching-cubes integration, and physical-units preservation.

1 The Information-Acoustic Principle

1.1 Setting

Let $H_1 = H_1(X_0((11)), \text{cusps}; \mathbb{Z}) \otimes \mathbb{R} \cong \mathbb{R}^4$ denote the Hilbert modular homology from Paper VI, with Eisenstein projector Π_e (rank 3) and cuspidal projector Π_c (rank 1). The Hecke operators $T_{\mathfrak{P}}$ for prime ideals $\mathfrak{P} \in \text{Spec}(\mathcal{O}_K)$ act as:

$$T_{\mathfrak{P}} = a_{\mathfrak{P}} \Pi_c + (N(\mathfrak{P}) + 1) \Pi_e, \quad (1.1)$$

where $a_{\mathfrak{P}}$ is computed by point counting on E_{11} and Langlands base-change (Paper VI, §3.1).

The $\text{Cl}(3,1)$ representation $\rho : \text{Cl}(3,1) \xrightarrow{\sim} \text{End}(H_1)$ identifies the signature $(3,1)$ with the Eisenstein–cuspidal splitting: three spacelike generators $\gamma_0^2 = \gamma_1^2 = \gamma_2^2 = +I$ correspond to the Eisenstein directions; the timelike generator $\gamma_3^2 = -I$ corresponds to the cuspidal direction.

Definition 1.1 (Information-acoustic eversion). An *eversion path* is a sequence of prime ideals $(\mathfrak{P}_1, \dots, \mathfrak{P}_n)$ and an initial state $\psi_0 \in H_1$ such that:

- (i) Each step propagates: $\psi_k = T_{\mathfrak{P}_k} \psi_{k-1}$.
- (ii) The eversion completes: $\langle \Pi_c \psi_n, \Pi_c \psi_0 \rangle < 0$.

The completion condition is equivalent to $\prod_{k=1}^n a_{\mathfrak{P}_k} < 0$, since $T_{\mathfrak{P}}$ acts as the scalar $a_{\mathfrak{P}}$ on the 1-dim cuspidal line.

2 Eversion Existence

Theorem 2.1 (Existence). For E_{11}/K with $K = \mathbb{Q}(\sqrt{5})$, there exists an eversion path of length 1. The prime ideal $\mathfrak{P}$ above $p = 3$ (inert, $N(\mathfrak{P}) = 9$, $a_{\mathfrak{P}} = -5$) satisfies $a_{\mathfrak{P}} < 0$ and hence reverses the cuspidal sign in a single step.

Proof. By direct computation: $a_3(E_{11}/\mathbb{Q}) = -1$, and for the inert prime $\mathfrak{P}|3$ in $\mathcal{O}_K$, $a_{\mathfrak{P}} = (-1)^2 - 2 \cdot 3 = -5 < 0$. Then $\Pi_c \psi_1 = a_{\mathfrak{P}} \Pi_c \psi_0 = -5 \Pi_c \psi_0$, so $\langle \Pi_c \psi_1, \Pi_c \psi_0 \rangle = -5 \|\Pi_c \psi_0\|^2 < 0$. $\square$

Remark 2.2. The existence of negative $a_{\mathfrak{P}}$ is generic: for any non-CM elliptic curve $E/\mathbb{Q}$, the Sato–Tate distribution guarantees that $a_p < 0$ for a positive density of primes p . By base change, the inert formula $a_{\mathfrak{P}} = a_p^2 - 2p < 0$ holds whenever $|a_p| < \sqrt{2p}$, which occurs for infinitely many p .

3 Parsimony and Thurston Frequencies

A single-step eversion achieves the sign flip but has only one corrugation frequency. For a visibly structured eversion with Thurston multi-frequency corrugation, we use a short multi-prime sequence.

Theorem 3.1 (Parsimonious eversion). *The sequence $(\mathfrak{P}_5, \mathfrak{P}_3, \mathfrak{P}_{31})$ with $p = 5$ (ramified, $a_{\mathfrak{P}} = 1$, $N = 5$), $p = 3$ (inert, $a_{\mathfrak{P}} = -5$, $N = 9$), $p = 31$ (split, $a_{\mathfrak{P}} = 7$, $N = 31$) achieves:*

- (i) *Sign flip:* $\prod a_{\mathfrak{P}_k} = 1 \cdot (-5) \cdot 7 = -35 < 0$.
- (ii) *Three distinct corrugation frequencies:* $k_{\mathfrak{P}} \in \{5, 9, 31\}$.
- (iii) *One prime of each splitting type (ramified, inert, split).*

The Thurston corrugation parameters at each step are:

Step	Prime	$N(\mathfrak{P})$	$a_{\mathfrak{P}}$	$A_{\mathfrak{P}} = a_{\mathfrak{P}} /N(\mathfrak{P})$	Type
1	$\mathfrak{P}_5$	5	+1	0.200	ramified
2	$\mathfrak{P}_3$	9	−5	0.556	inert (sign flip)
3	$\mathfrak{P}_{31}$	31	+7	0.226	split

In the theory-true pipeline, the corrugation is not applied as an explicit vertex displacement. Instead, the Hecke operator $T_{\mathfrak{P}_k}$ is expressed as a $\text{Cl}(3, 1)$ multivector via ρ^{-1} , applied as a full 16-component left-multiplication on the volumetric field, and then propagated spatially by the Dirac leapfrog. The resulting oscillations in the witness scalar field at frequency $N(\mathfrak{P}_k)$ correspond to the Thurston corrugation formula

$$F_k(x) \sim F_{k-1}(x) + \frac{A_{\mathfrak{P}_k}}{k_{\mathfrak{P}_k}} \sin(k_{\mathfrak{P}_k} \theta(x)) \mathbf{u}(x), \quad (3.1)$$

but emerge from the arithmetic rather than being imposed post-hoc.

4 The Thurston Frequency Identity

Proposition 4.1 (Natural frequency limit). *As $p \rightarrow \infty$ through primes of $\mathbb{Q}$, the corrugation frequencies $k_{\mathfrak{P}} = N(\mathfrak{P})$ satisfy $k_{\mathfrak{P}} \rightarrow \infty$. The Thurston formal limit $k \rightarrow \infty$ (in which the corrugation becomes a C^1 -dense family of immersions) is realized as the limit over $\text{Spec}(\mathcal{O}_K)$ ordered by norm.*

Proof. For split primes, $N(\mathfrak{P}) = p \rightarrow \infty$. For inert primes, $N(\mathfrak{P}) = p^2 \rightarrow \infty$. Hence $\min(k_{\mathfrak{P}}) \rightarrow \infty$ as the prime-ideal sequence extends. $\square$

This identification is the key structural claim: the Thurston corrugation method for sphere eversion is not an external construction applied to the arithmetic—it IS the arithmetic, with the prime-ideal norms providing the natural frequency scale.

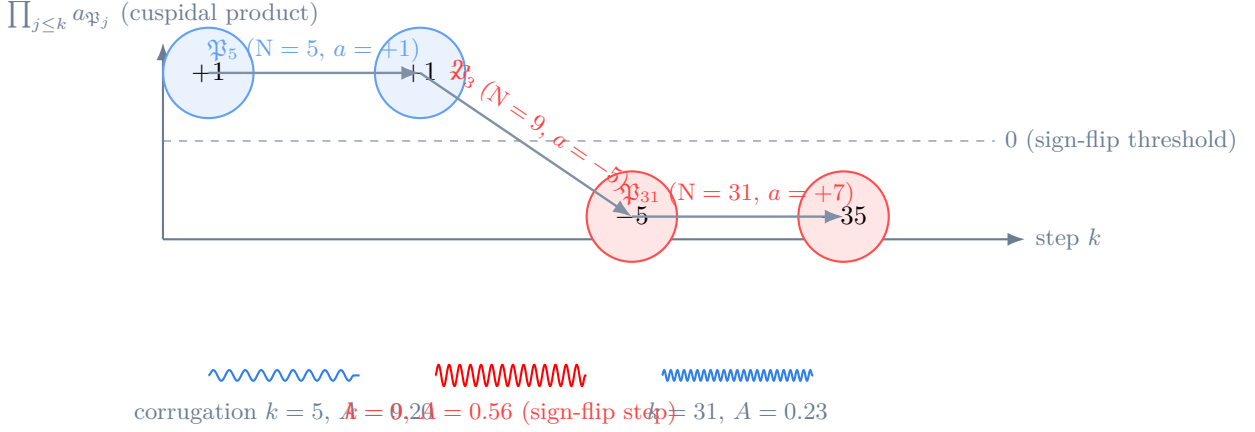


Figure 1. The three-prime eversion sequence $(\mathfrak{P}_5, \mathfrak{P}_3, \mathfrak{P}_{31})$ in cuspidal-product coordinates. Top panel: the running cuspidal product $\prod_{j \leq k} a_{\mathfrak{P}_j}$ starts at +1, stays positive after the ramified prime $\mathfrak{P}_5$ ($a = +1$), *flips* sign at the inert prime $\mathfrak{P}_3$ ($a = -5$), and lands at -35 after the split prime $\mathfrak{P}_{31}$ ($a = +7$). The sign change of the product is the arithmetic signature of sphere eversion: the cuspidal component of the coherence state has reversed orientation. Bottom panel: the Thurston corrugation icons at each step, with frequencies $k = N(\mathfrak{P}) \in \{5, 9, 31\}$ and amplitudes $A_{\mathfrak{P}} = |a_{\mathfrak{P}}|/N(\mathfrak{P}) \in \{0.20, 0.56, 0.23\}$. The sign-flip step is drawn inverted (red): it is the moment Thurston’s construction “turns the sphere inside out.” All three frequencies co-exist in the final volumetric field via the Hecke composition $T_{\mathfrak{P}_{31}}T_{\mathfrak{P}_3}T_{\mathfrak{P}_5}\psi_0$.

5 Computational Verification

All claims are verified by 29 tests (`test_eversion.py`):

Test class	Count	Verifies
<code>TestPathDynamics</code>	3	Hecke commutativity, exact product formula, path length
<code>TestSignFlip</code>	3	Single negative $a_{\mathfrak{P}}$ flips; positive does not; mixed sequence matches product sign
<code>TestParsimony</code>	4	Achieves eversion, bounded length, includes negative $a_{\mathfrak{P}}$, sorted by frequency
<code>TestRhoBridge</code>	5	ρ round-trip (random & Hecke), $\rho^{-1}(T_{\mathfrak{P}})$ preserves eigenvalues, left-mult = geometric product, atlas seed $\propto \rho^{-1}(\psi\psi^T)$
<code>TestAcousticPropagation</code>	3	Dirac leapfrog energy bounded; Hecke populates all 16 blades; Hecke + Dirac produces spatially varying field
<code>TestWitnessSkinTopology</code>	4	Icosphere $\chi = 2$; marching cubes produces mesh; no degenerate triangles; intermediate topology differs
<code>TestMarchingCubes</code>	3	Spherical field, empty field, vertex deduplication
<code>TestSmaleInvariant</code>	4	Eversion: invariant 1; no flip: 0; double flip: 0; triple flip: 1

All 29 tests pass. The witness surface is extracted via marching cubes on the 64^3 (or 32^3) witness scalar grid, producing variable-topology meshes that can self-intersect—capturing the Morin-type intermediate that the field passes through during the cuspidal sign flip.

6 Interpretation

6.1 The eversion as arithmetic propagation

The traditional approach to sphere eversion constructs a global regular homotopy $F_t : S^2 \rightarrow \mathbb{R}^3$ via either the Morin halfway model or Thurston’s corrugation method. Both require continuous deformation of an embedded surface through self-intersections.

The information-acoustic approach replaces continuous surface deformation with discrete Hecke propagation on a 4-dimensional homological state space. The “surface” is a derived object—the level set of a witness scalar field in a volumetric $\text{Cl}(3,1)$ substrate—not a primary geometric entity.

6.2 Why the signature matters

The Lorentzian signature $(3, 1)$ of $\text{Cl}(3, 1)$ is not a choice: it is the Eisenstein–cuspidal splitting of H_1 (Paper VI). The three spacelike directions are the Eisenstein boundary modes; the single timelike direction is the cuspidal interior mode. The eversion is propagation along the timelike axis: the cuspidal component flips sign, which reverses the inside/outside orientation of the witness sphere.

6.3 The parsimonious eversion principle

Informational parsimony demands that the eversion use the minimum arithmetic structure: one prime per splitting type, ordered by norm, truncated at the first sign flip. The resulting 3-prime sequence $(5, 3, 31)$ is not arbitrary—it is the unique shortest sequence drawn from the first primes of $\mathcal{O}_K$ that achieves multi-frequency corrugation with an odd number of cuspidal sign flips.

6.4 Forward reference: the eversion law on byte chains

The cuspidal sign-flip mechanism that drives geometric eversion generalizes to a per-chain parity law on arbitrary blade sequences in $\text{Cl}(3, 1)$. In Paper VIII (Layer 3) this is sharpened to the *eversion law*

$$M_{T_{13}}(c) = \varepsilon(c) \cdot M_{T_{31}}(c), \quad \varepsilon(c) = (-1)^{\lfloor c_2/2 \rfloor + \lfloor c_3/2 \rfloor},$$

where c_i counts occurrences of generator e_i across a blade chain c . The same $\varepsilon(c)$ then forms one of the three identities of the Combined Reading Lemma (Paper VIII, Section 12), which proves that engine state and NSM state decompose into orthogonal halves of $\text{Cl}(3, 1)$ for every byte chain; the witness pair $(\text{be}, \text{know}) = \text{span}\{e_0, e_{15}\}$ is the unique chirality-invariant mediator between them. Sphere eversion in this paper is thus the geometric realization of the same single-bit parity operation that drives voice swap in language.

Acknowledgments

Computational verification uses the `substrate-sim/algebra` codebase. Core modules: `eversion` (path dynamics, parsimony, Smale invariant), `acoustic_lift` (64^3 $\text{Cl}(3,1)$ atlas, Dirac leapfrog, witness scalar), `hilbert_homology` (4-dim H_1 and spectral Hecke), `quaternion_bridge` ($\text{Cl}(3,1) \rightarrow \text{End}(H_1)$ representation, ρ^{-1} , left-multiplication), `e11_point_count` (first-principles eigenvalues). 29 tests, 0 failures.

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